Spatial is Special

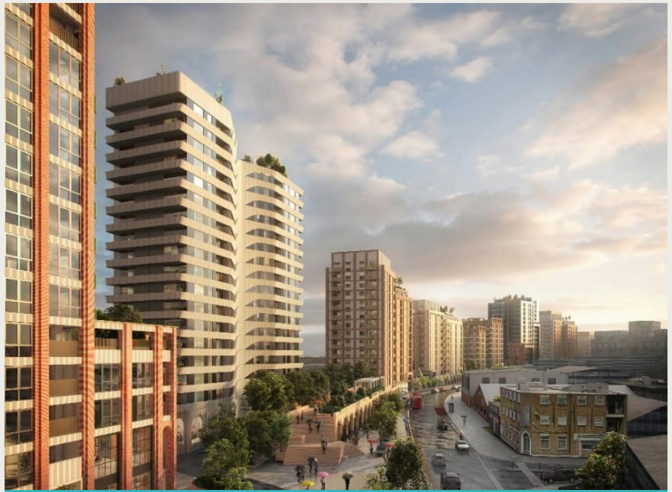
How We Use Geospatial Data Science To Make An Impact

5 September 2024

Who are we?

Engineering.
Management.
Development.

Advisory



Buildings and Cities

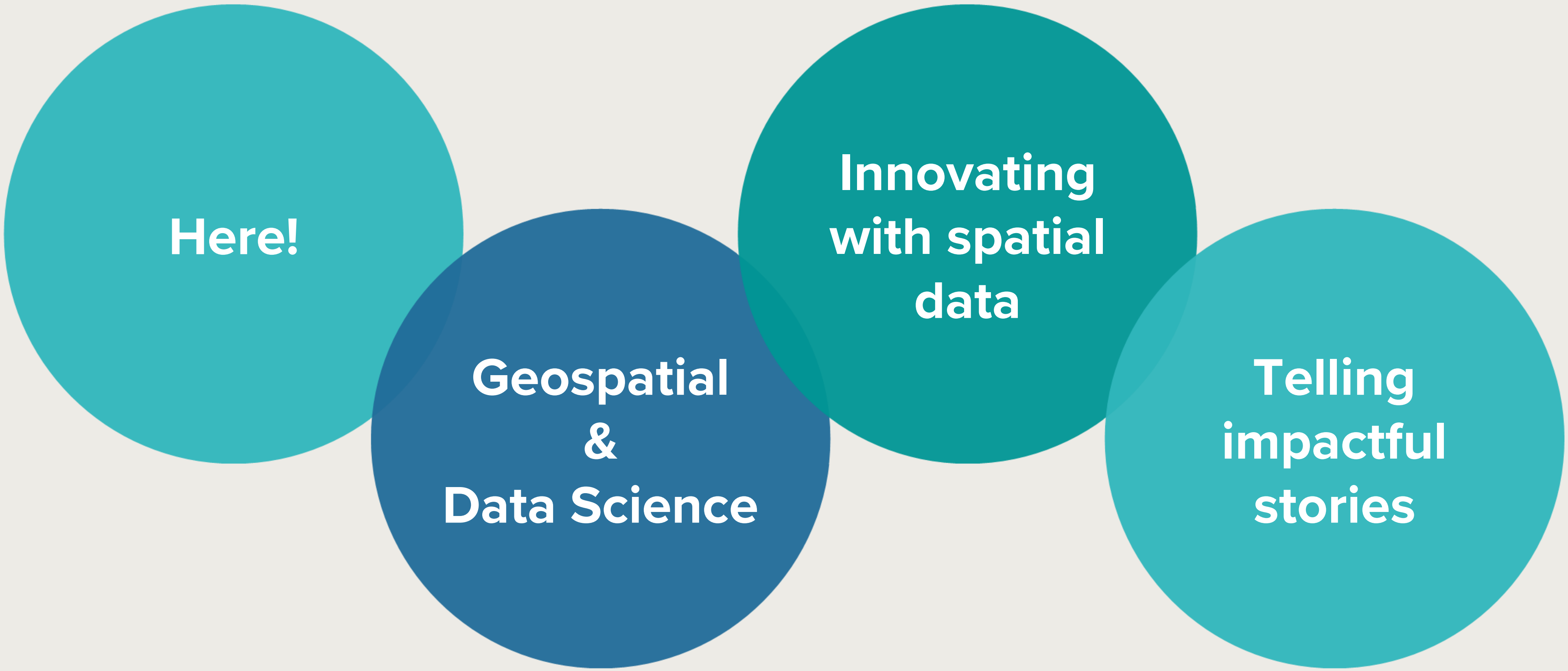
Energy

Environment

Transport

Water

My Background


Here!
Geospatial & Data Science
Innovating with spatial data
Telling impactful stories

Humans

love

storytelling

My Story!

1. Life as a Starbies Barbie
2. Finding that *Spatial* Someone
3. Are Trams for Tramps? Trolleying through hidden insights…
4. Takeaways

Life as a

Starbies Barbie


MATTHEW

Cities

City-wide Energy Masterplanning

Aims

- Identifying credible opportunities to decarbonise
- Baselining energy performance of housing stock

Angles

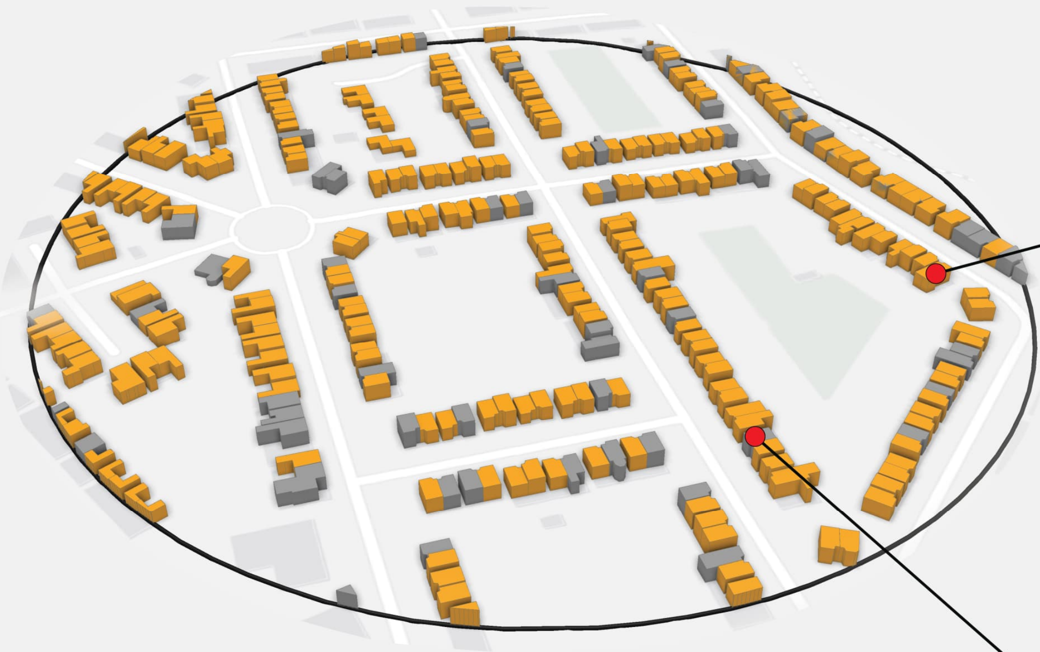
- What does "the current" look like?
- How do we deal with missing data?
- What makes a place unique?

Approach

- Statistical imputation methods
- Geodemographic Classification for pen-portraits and characterisation

What's Here? What's Missing?

Imputation of EPC data



Complete Dataset

Current EPC:	F
Potential EPC:	D
Age:	65-75
Height:	9m
Surface Area:	$57m^2$
Street Address:	178 Park Road
Output Area:	E0048704

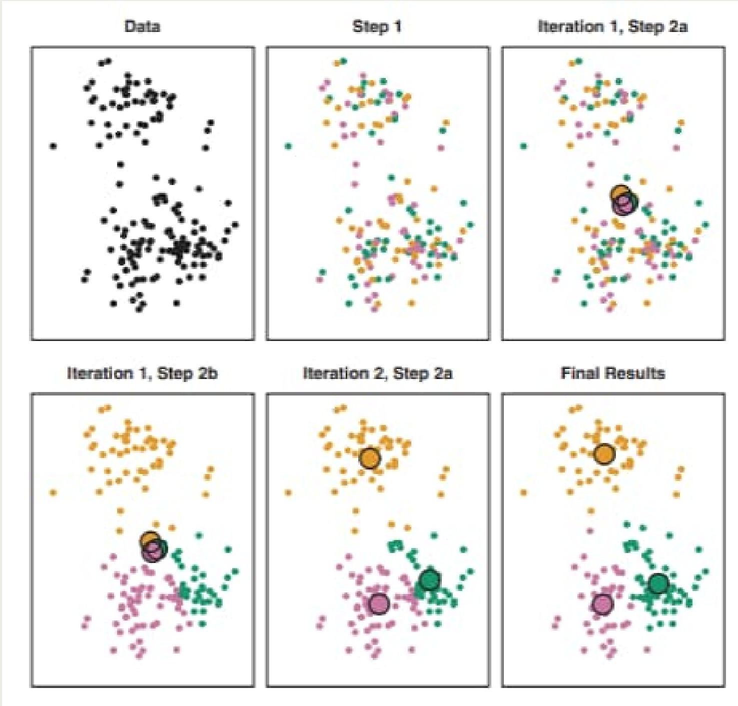
Incomplete Dataset

Current EPC:	?
Potential EPC:	?
Age:	85+
Height:	15m
Surface Area:	$69m^2$
Street Address:	5 Park Grove
Output Area:	E0048704

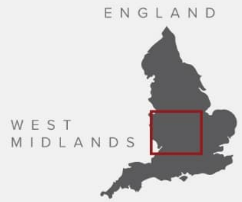
1. K-means to identify clusters
2. Nine underlying variables
3. Differences of each cluster against the global mean make it “unique”

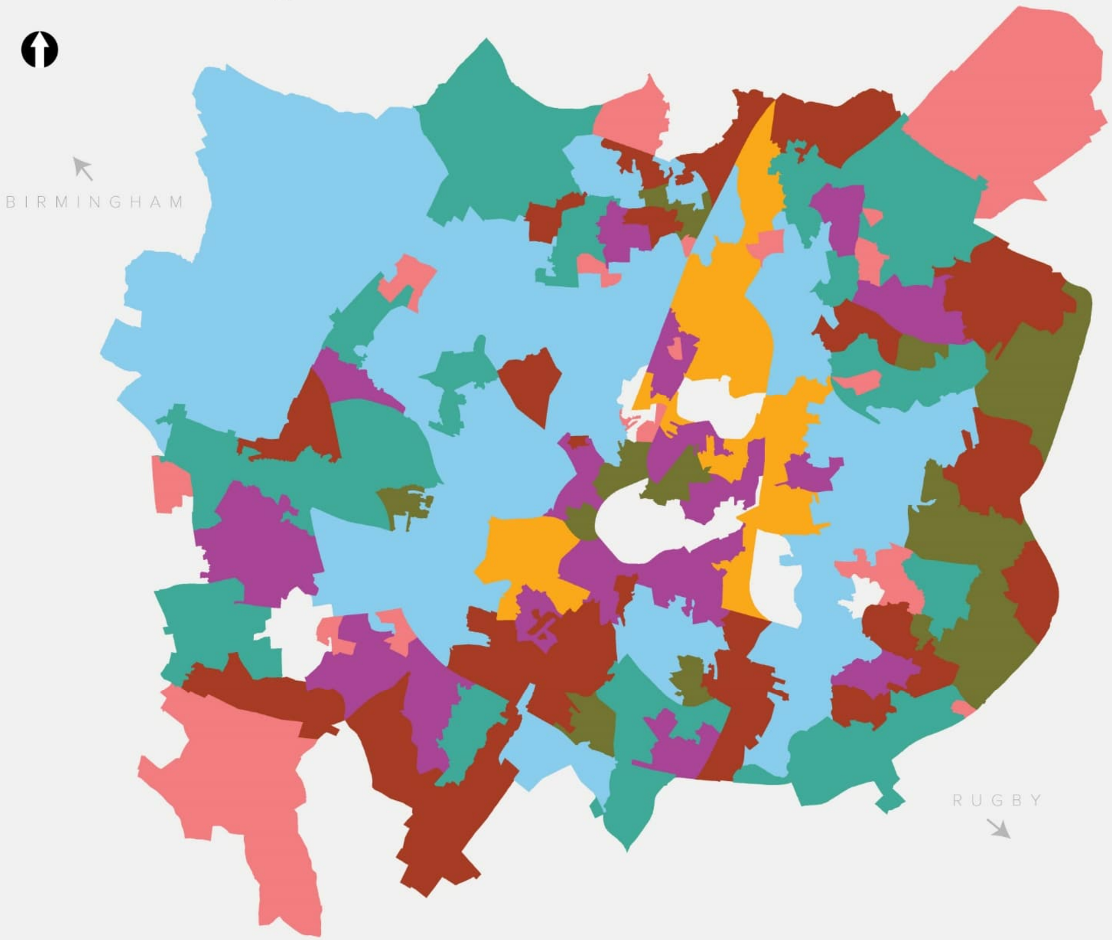
Algorithm 10.1 *K-Means Clustering*

1. Randomly assign a number, from 1 to K, to each of the observations. These serve as initial cluster assignments for the observations.
2. Iterate until the cluster assignments stop changing:
   (a) For each of the K clusters, compute the cluster *centroid*. The kth cluster centroid is the vector of the p feature means for the observations in the kth cluster.
   (b) Assign each observation to the cluster whose centroid is closest (where *closest* is defined using Euclidean distance).



Describing The EPC Rating Makeup Of A City





Cluster 1
Mix of detached houses and flats aged 60 to 69 years, with a current EPC Rating of "B" and potential EPC Rating of "A"

Cluster 2
Mix of terraced houses and flats aged 51 to 124 years, with a current EPC Rating of "D" and potential EPC Rating of "C"

Cluster 3
Terraced houses aged greater than 124 years, with a current EPC Rating of "G" and potential EPC Rating of "F"

Cluster 4
Semi-detached houses aged between 60-69 years with a current EPC Rating of "B" and potential EPC Rating of "A"

Cluster 5
Detached houses aged between 60 to 69 years with a current EPC rating of "B" and potential EPC Rating of "A"

Cluster 6
Mix of detached and semi-detached houses aged between 41 and 51 years with a current EPC Rating of "B" and potential EPC Rating of "B"

Cluster 7
Mix of HMOs and Terraced Houses aged between 106 and 124 years with a current EPC Rating of "B"

Quantitative
compliments
qualitative
approaches

Finding that *Spatial* Someone?

The date she wants

The date I can afford

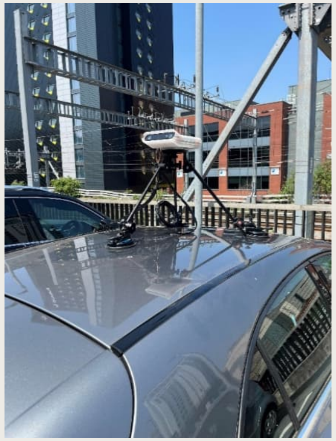
```
> # Using POSIXct
> timect <- as.POSIXct("2022-11-08 22:14:35 PST")
> print(timect)
[1] "2022-11-08 22:14:35 PST"
> class(timect)
[1] "POSIXct" "POSIXt"
```

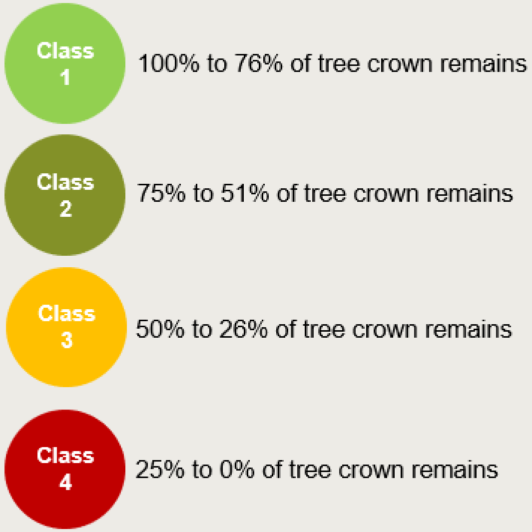
Environment

Ash Dieback Risk Model

Aims

- Increasingly prominent problem for councils
- Ash Dieback Trees on public footways / highways
- Understanding “the where”




Class
1
100% to 76% of tree crown remains
Class
2
75% to 51% of tree crown remains
Class
3
50% to 26% of tree crown remains
Class
4
25% to 0% of tree crown remains

Aims

- Increasingly prominent problem for councils
- Ash Dieback Trees on public footways / highways
- Understanding “the where”

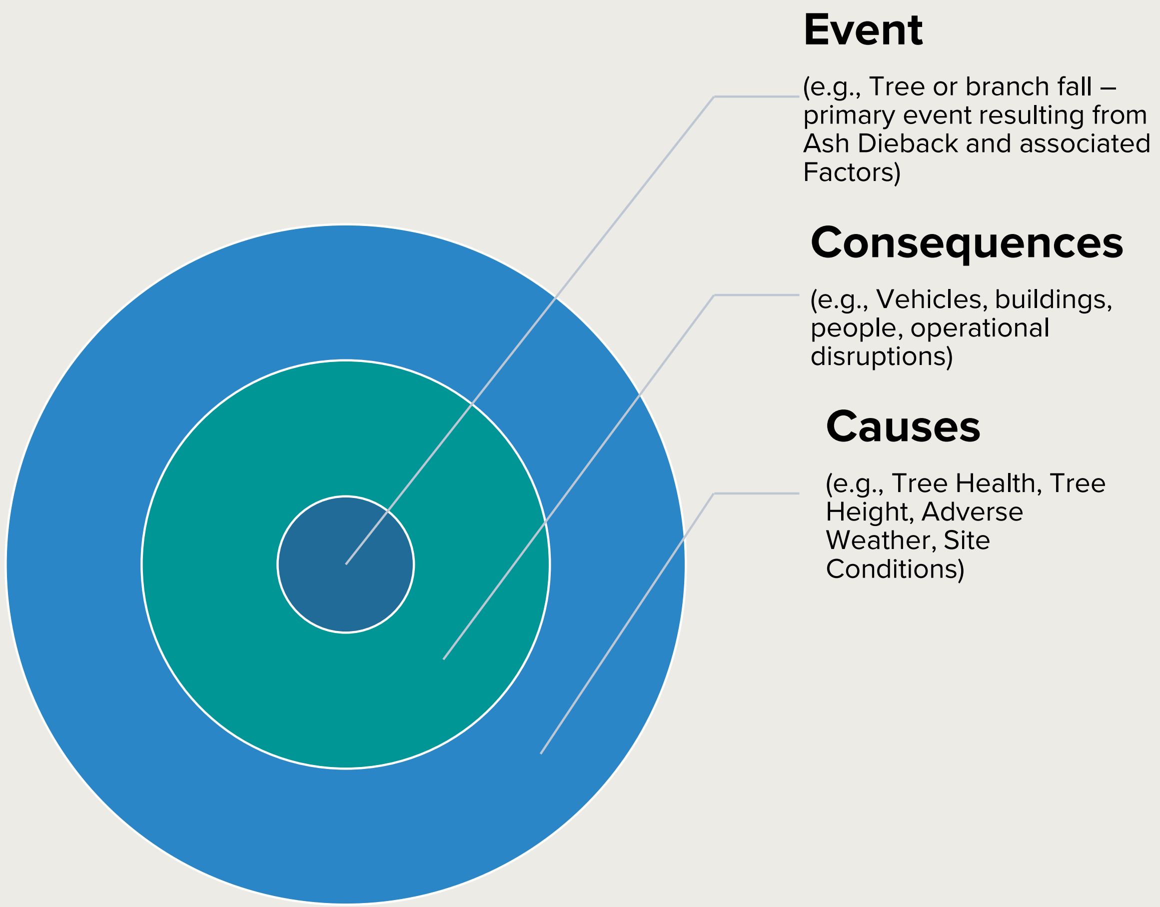
Angles

- How do we define risk?
- What other data is available?
- How do we weight the variables?

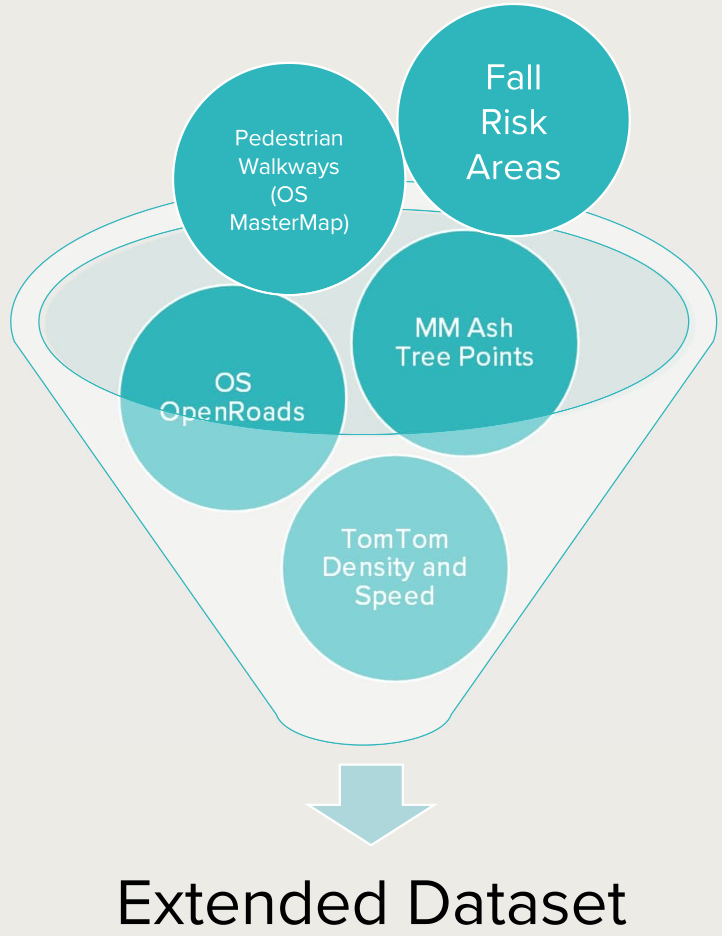
Approach

- Industry guidance
- Consultation of SMEs
- Edge cases?

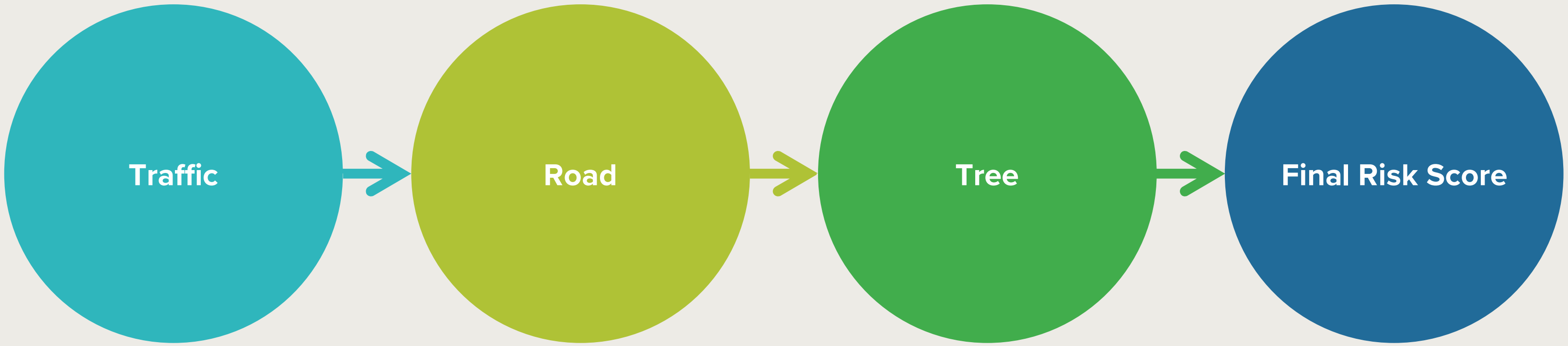
Risk Scores



Event
(e.g., Tree or branch fall – primary event resulting from Ash Dieback and associated Factors)
Consequences
(e.g., Vehicles, buildings, people, operational disruptions)
Causes
(e.g., Tree Health, Tree Height, Adverse Weather, Site Conditions)



Pedestrian Walkways (OS MasterMap)
Fall Risk Areas
OS OpenRoads
MM Ash Tree Points
TomTom Density and Speed
Extended Dataset



What's around?

- Traffic Speed
- Traffic Density

Where is it?

- Pedestrianisation
- Type of Road

How severe?

- Disease stage
- Tree height
- Fall zone

Everything combined

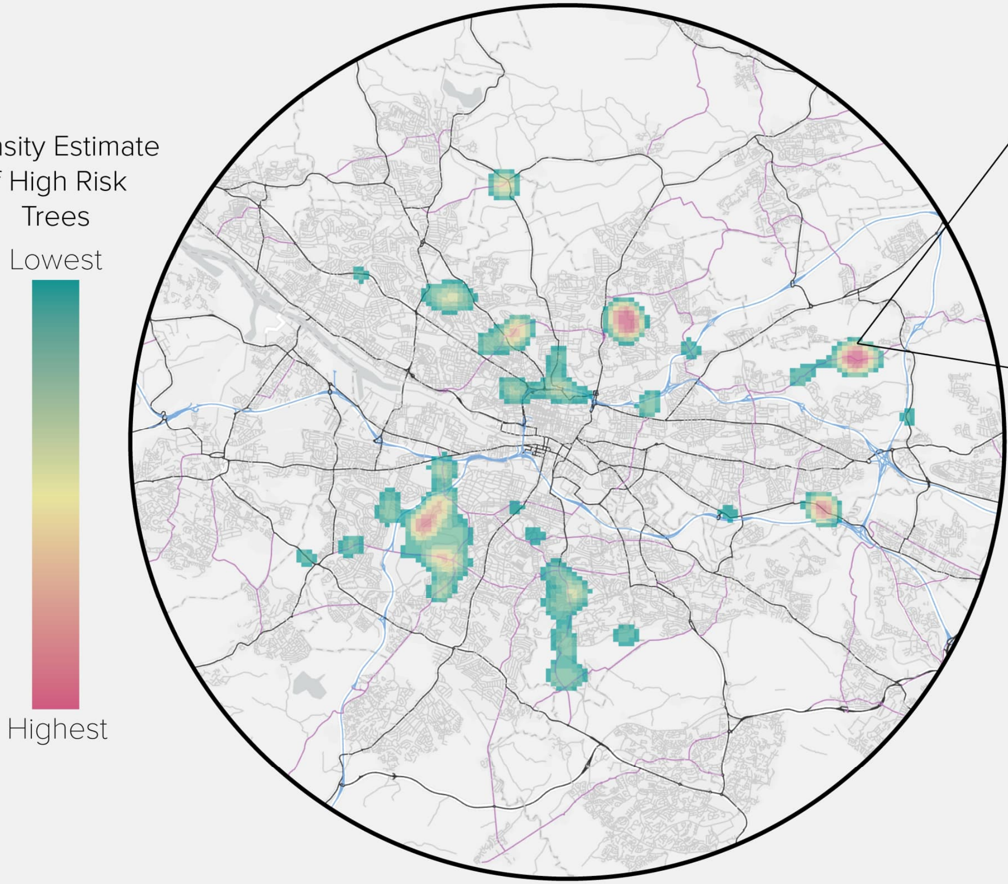
- Scored 1-10
- Weighted combination

New Relationships, Patterns and Insights

Density Estimate of High Risk Trees

Lowest

Highest

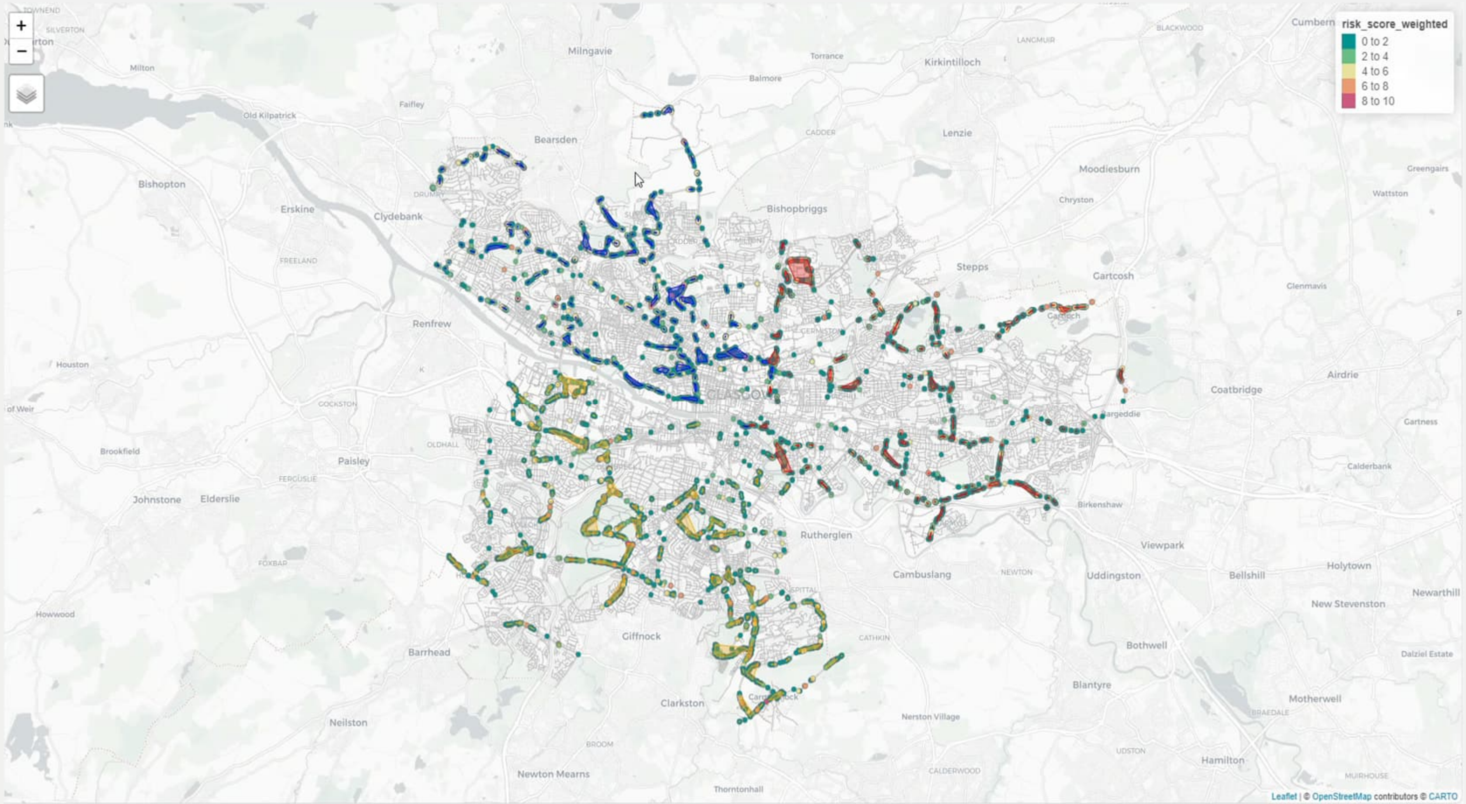


Tree 4326

Traffic Density:	5
Traffic Speed:	4
Fall Area:	3
Pedestrianisation:	2
Dieback Class:	4

Weighted Risk Score: 5 (High Risk)

Towards More Strategic Decision Making



Uncovering spatial **relationships** in data consolidates decisions

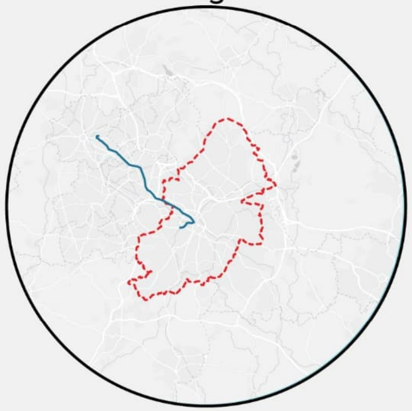
Are Trams for Tramps? Trolleying through hidden insights…


Birmingham
easyGym
21

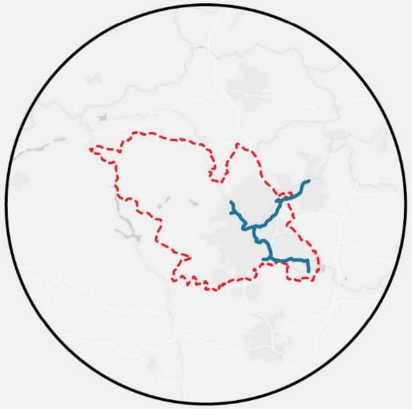
What Can We Notice?

Tram lines in 6 Cities

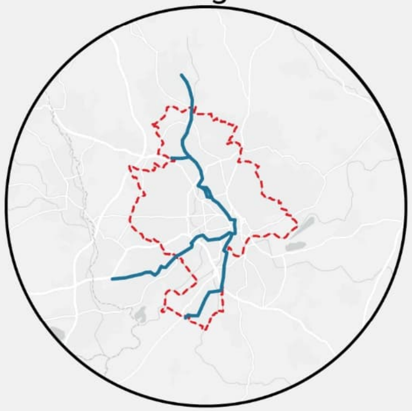
Birmingham



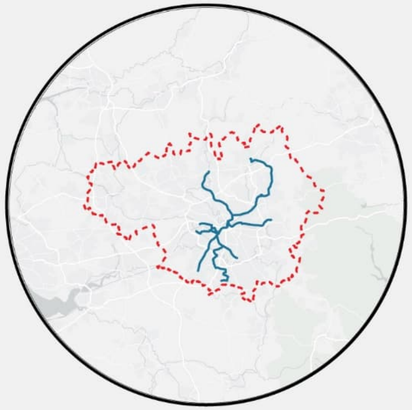
Sheffield



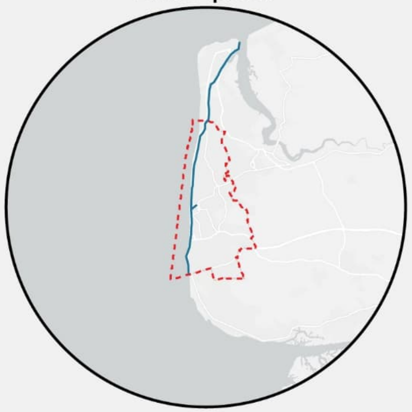
Nottingham



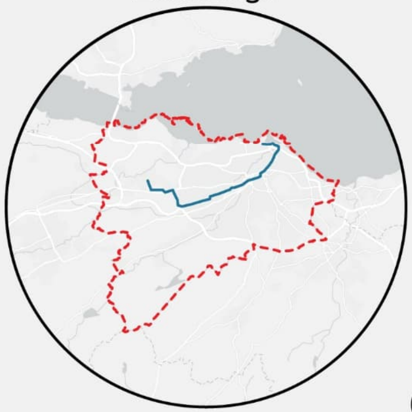
Manchester



Blackpool

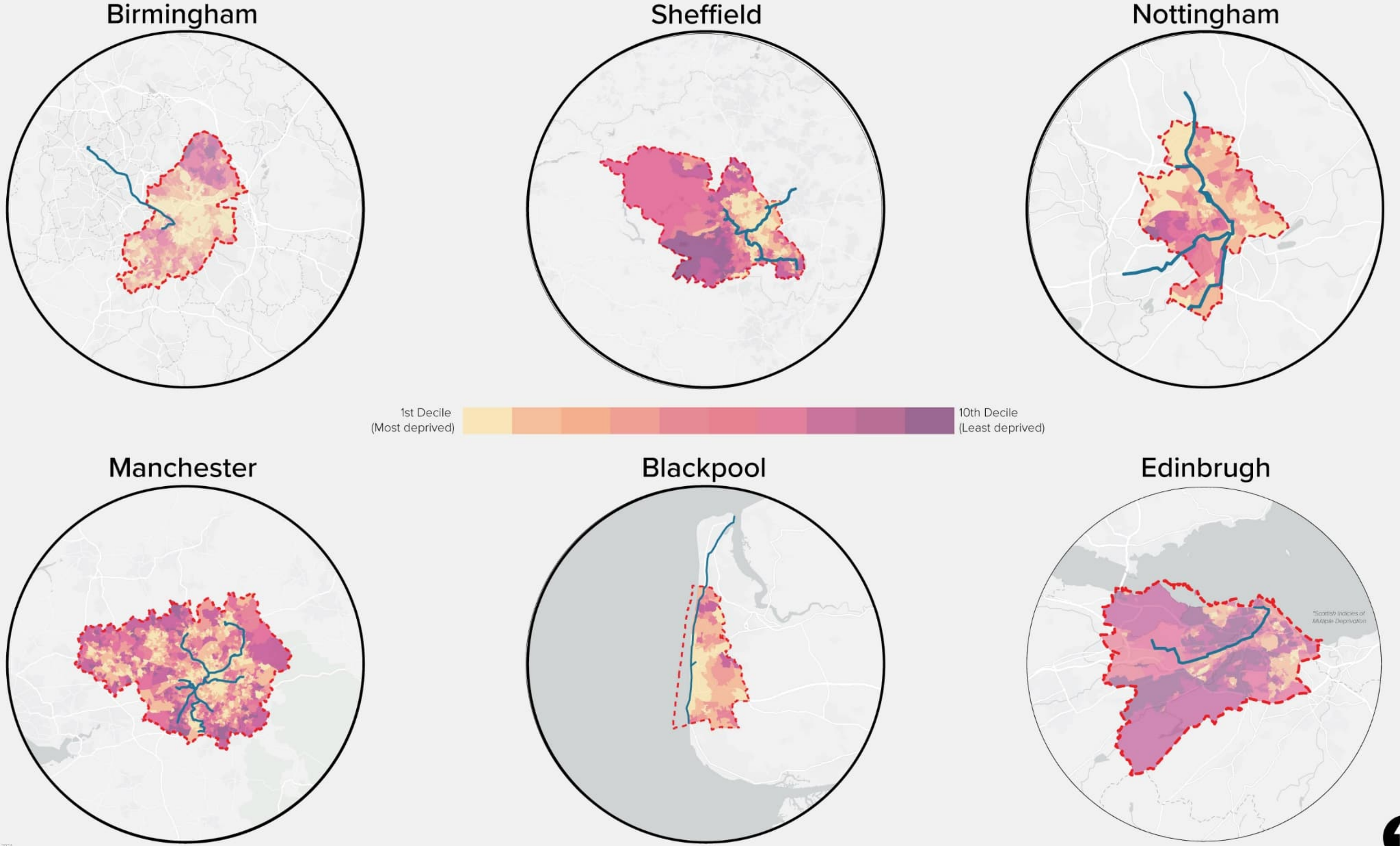


Edinburgh



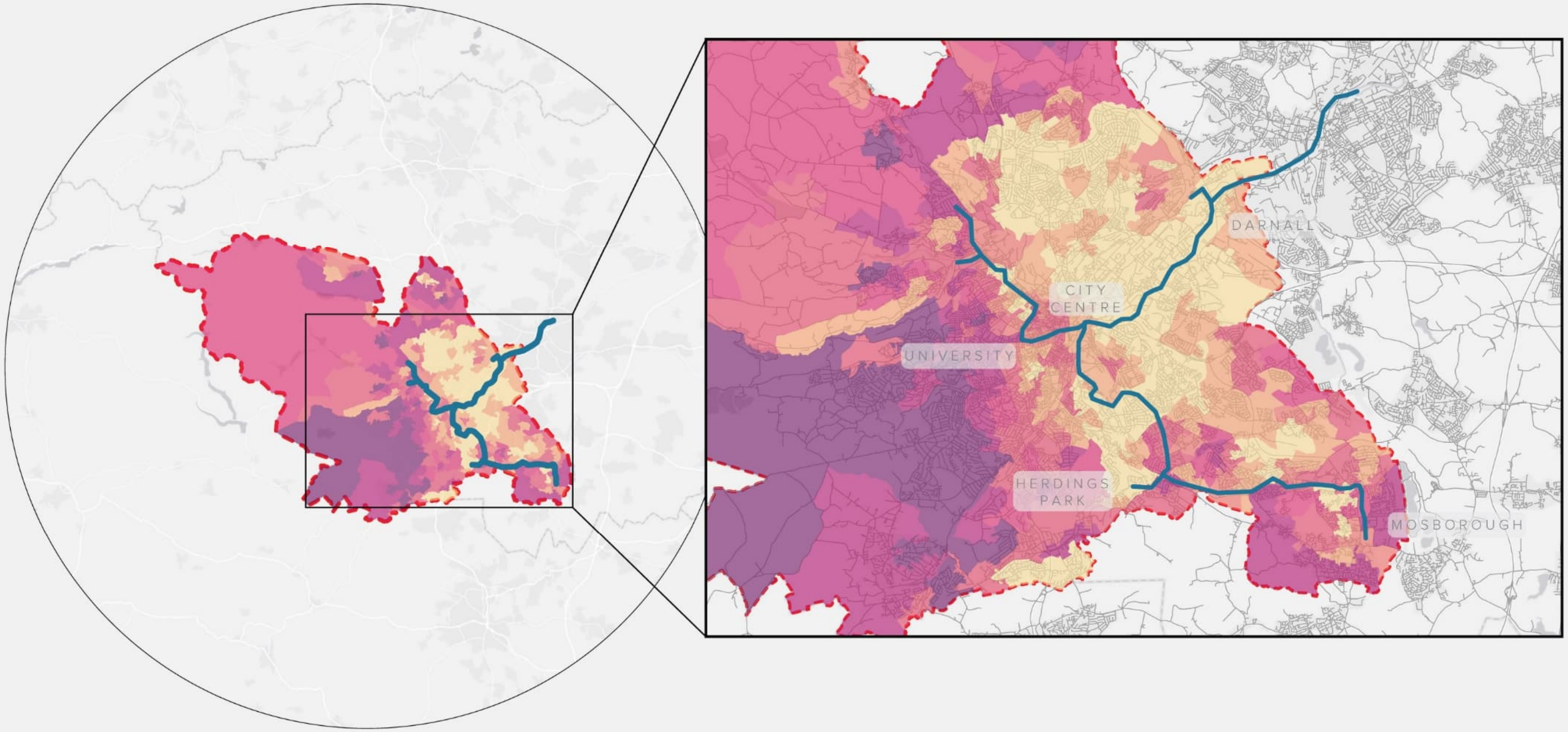
Can We Dig Deeper?

Tram lines in 6 Cities



Trolleying Through Insights...

Deprivation along Sheffield's Tram Network



Net-Zero

Net Zero Neighbourhoods Masterplanning

Aims

- Holistic place-based approach to net-zero
- Baselining and context-setting
- Provide recommendations for interventions

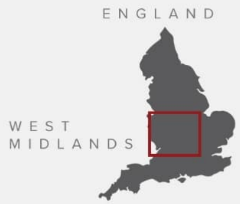
Angles

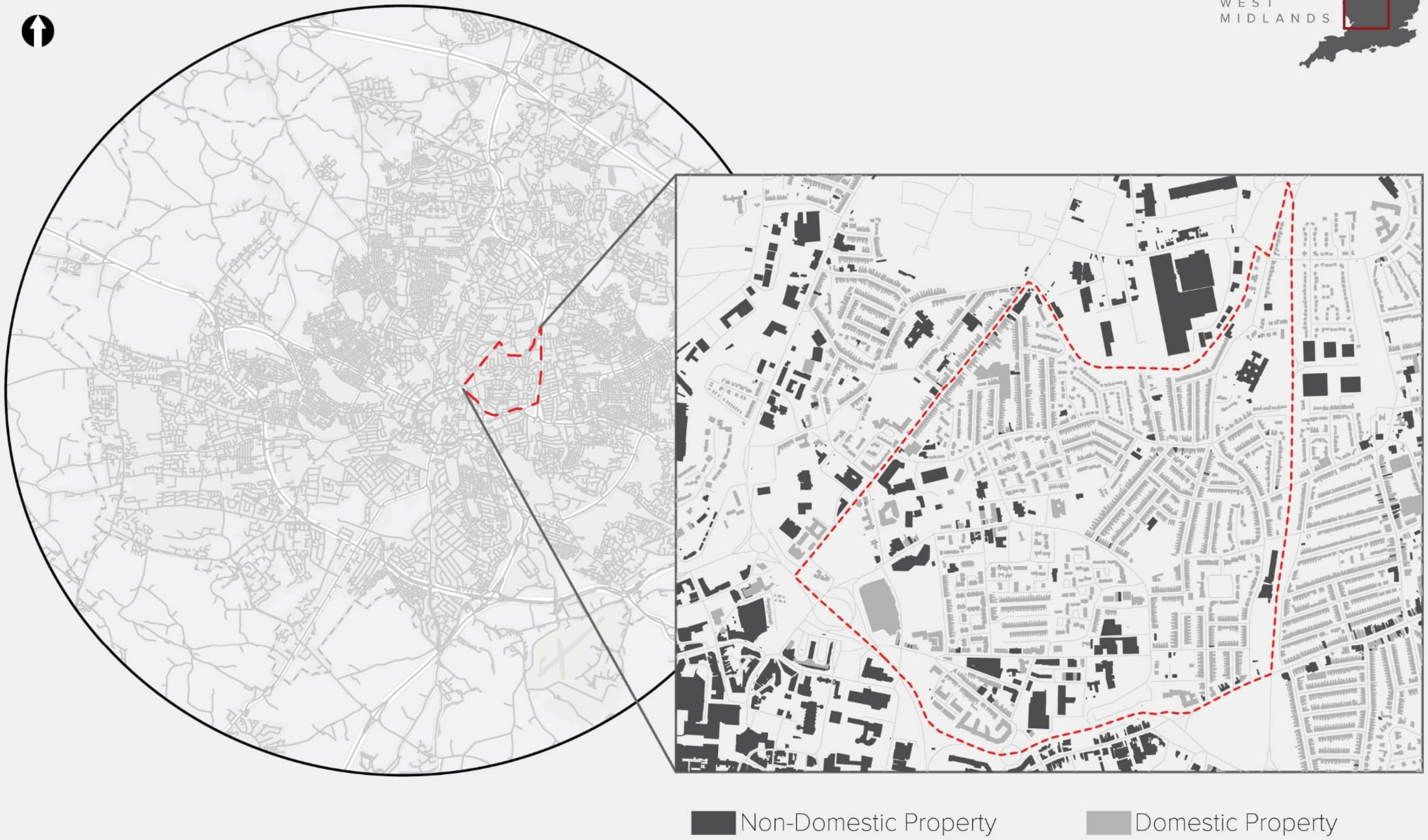
- What's already there?
- How do we measure abstract concepts?
- Can we synthesize to reveal patterns?

Approach

- Data from OpenStreetMap
- Mining walk times using Open Source Routing Machine (OSRM)
- Construct a composite index

Towards a Net Zero Neighbourhood



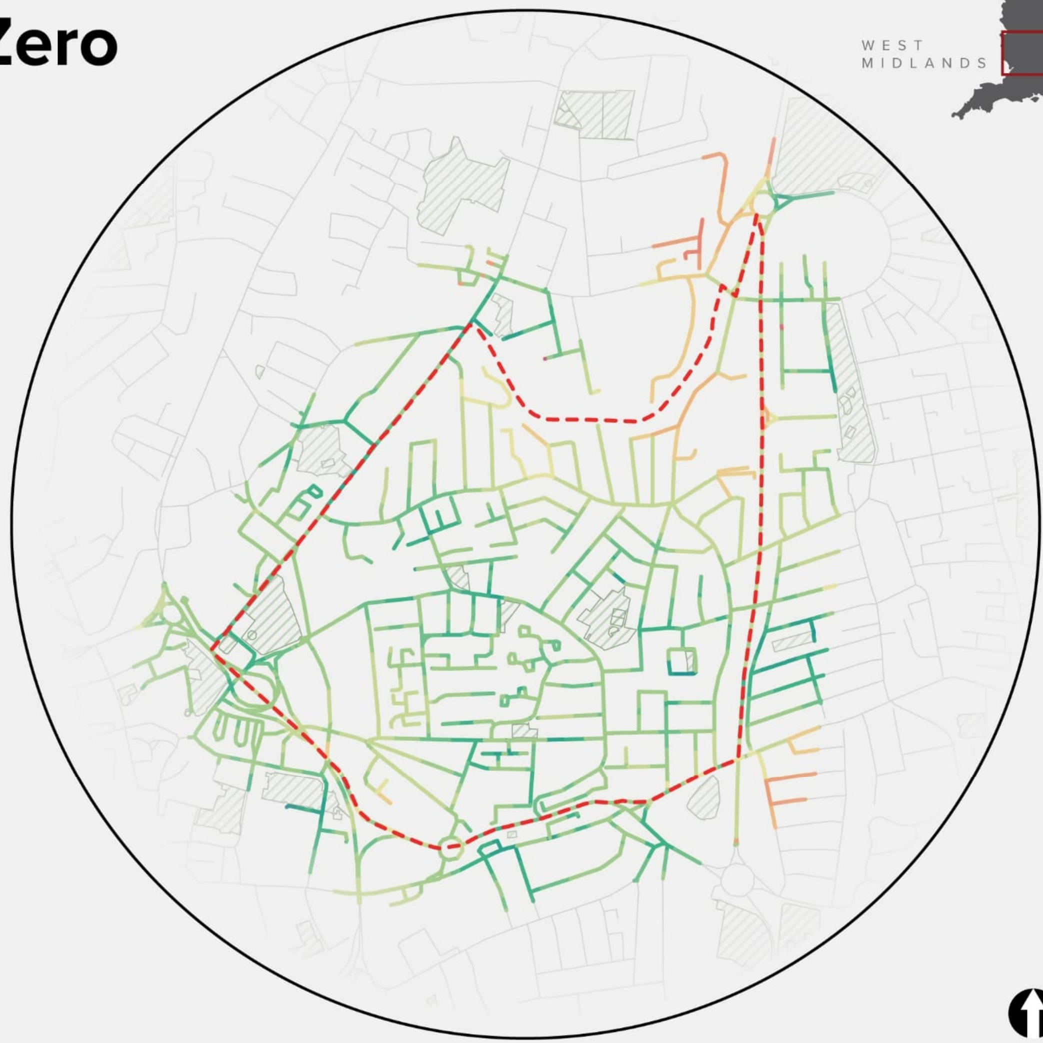


Walkability of Greenspaces Within the Net Zero Neighbourhood

Walk time to nearest green space (mins)

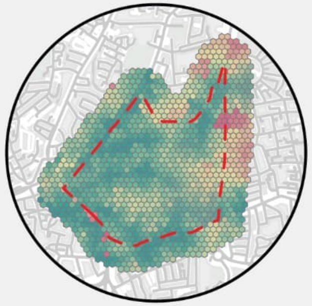
- 0-1
- 1-2
- 2-3
- 3-4
- 4-5
- 5-6
- 6-7
- 7-8
- 8-9
- 9-10
- 10-11
- 11-12
- 12-13

Green space

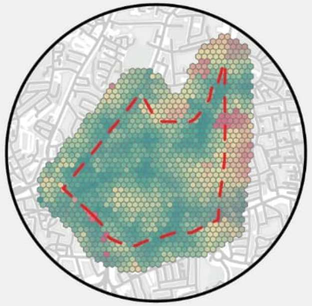


Measuring Amenities Accessibility

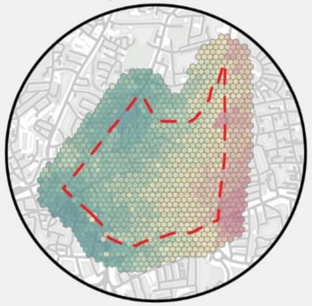
Shops



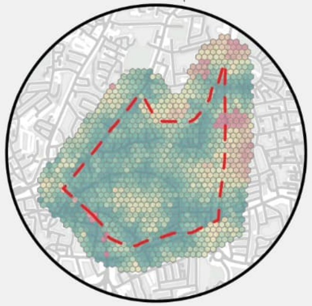
Schools



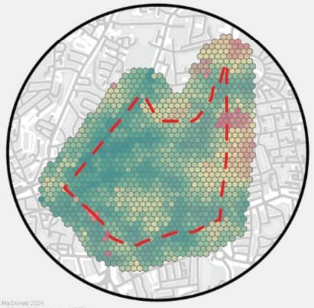
Supermarkets



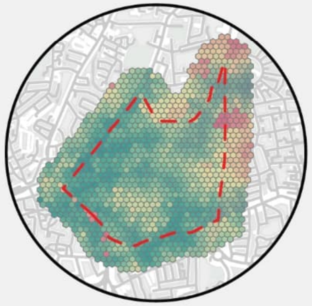
Bus Stops



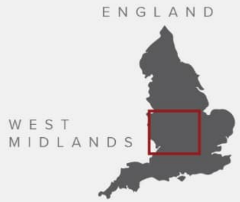
GPs



Pharmacies



Amenities Accessibility Composite Index



Accessibility Score

Best Accessibility

- 0-1
- 1-2
- 2-3
- 3-4
- 4-5
- 5-6
- 6-7
- 7-8
- 8-9
- 9-10

Worst Accessibility

Digging deeper
helps us **see**
hidden patterns

Takeaways

1 Spatial is Special, Spatial is Sexy

2 Ask Spatially Special Questions

3 Match made in heaven?

4 Now back to you...

What makes spatial special to **you**?

Thank you

5 September 2024

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